Jacob Pomeranz

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EDUCATION

University of Virginia — *Bachelor's in Computer Science*

Aug 2014 – May 2017

Rodman/Echols Scholar. GPA: 4.0

EXPERIENCE

Senior Software Engineer, *Qualtrics*

Aug 2017 – Present

- Performed a zero-downtime migration of all xFlow services from AWS SWF to Temporal as an orchestration backend across 6 services. Saved 20% of Temporal costs by building a replacement for activity heartbeating
- Re-architected asynchronous workflow platform and adopted Kafka as a unified event bus to scale to support tens of millions of executions per day. Created a rate-limiting and traffic-shaping solution for workflows that enabled 100x scale increase in throughput
- Migrated the workflow runtime to a graph data model adding support for branching, looping, and advanced control logic
- Added Mutual TLS authentication support to the Qualtrics Public API
- Added default E2E encryption alongside BYOK encryption support to 8 services
- Resolved an org-wide gap in API documentation by creating the Internal API Catalog, which went on to host OpenAPI docs for over 209 Qualtrics services and 73 teams
- Drove a Public API documentation migration between vendors which tied API docs to the source of truth by using OpenAPI specs and launched the new documentation site for customers. Held vendors, technical writer contractors, support, and partner teams accountable for documenting 232 Public API endpoints by hard deadlines. Organized bug bashes with engineering, support, and resolution teams in addition to writing 19 new OpenAPI specs
- Created a supplemental data microservice enabling caching and global replication of arbitrary structured data in Postgres with dynamic indexes within the survey-taking experience at low latency. Enabled scenarios such as indexed lookups and an autocomplete question surveys with robust usage by over 72 customer brands within 1 year of release
- Used NodeJS, Golang, and Java in conjunction with MySQL, DynamoDB, and Redis to develop new microservices along with React to develop new user interfaces
- Built backend infrastructure using Kubernetes, Nomad, Docker, Terraform, and AWS CDK

Computer Science Intern, *National Security Agency*

May 2016 – Aug 2016

- Underwent Agency Special Background Investigation and polygraph testing to hold TOP SECRET/SI/TK clearances
- Used Python, SQLite, and PostgreSQL to perform large-scale extraction, processing, and transformation of critical data
- Built several key database visualization tools using PHP and Ajax
- Restructured development environment and automated tasks to increase efficiency and stability

Web Development Intern, *OpenGrounds at UVA*

Mar 2015 – Sep 2015

- Worked with a student design team and used technologies such as Python, Django, SQLite, and Bootstrap to create a novel research website for word-cloud generation

New Horizons GSST Mentorship, *Thomas Jefferson National Accelerator Facility*

Aug 2013 – May 2014

- Worked with Dr. David Lawrence to create an improved subatomic charged particle tracking program using C++ and Nvidia CUDA for the GlueX experiment
- Presented my research to receive the GSST Outstanding Research award

US Department of Energy Internship, *Thomas Jefferson National Accelerator Facility*

Jun 2013 – Jul 2013

- Independently set up an open-source monitoring and management infrastructure (Nagios)
- Coded various plugins and scripts using C and Bash. Helped with Windows XP EOL migration. Installed and managed computer hardware and servers throughout the lab